BEE Lab Project Report

Retail Sales Analytics: Excel & Power BI Dashboards

$334,395	$51,612	15.4%	300
Total Sales	Total Profit	Profit Margin	Orders

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Problem Statement & Objectives

"Retail businesses generate large volumes of transactional data but often lack tools to extract insights. This project builds two interactive BI dashboards — in Excel and Power BI — enabling stakeholders to monitor KPIs, track trends, and forecast sales without programming knowledge."

Objectives
Build Excel Dashboard with Pivot Tables, PivotCharts & Slicers
Build Power BI Dashboard with 16 DAX measures & 3-month Forecast
Perform 3-Year YoY & MoM Sales Trend Analysis (2020-2022)
Identify top products, loss-making sub-categories, best regions

Dataset Overview

300 Orders
Total Records

3 Years
2020 / 2021 / 2022

14 Columns
Full Attribute Set

3 Categories
Tech | Furniture | Office

4 Regions
E | W | Central | S

21 Losses
High-discount orders

File: retail_sales_final.csv | Source: Kaggle Superstore Dataset

Category Split: Technology $250,980 (75%) | Furniture $74,164 (22%) | Office Supplies $9,251 (3%)
Region Split: West $101,195 | South $84,679 | East $81,436 | Central $67,086
Segment Split: Consumer $159,089 (48%) | Corporate $101,812 (30%) | Home Office $73,494 (22%)

Section 2: Excel Dashboard — Store Sales Performance

DASHBOARD

KPI cards (10 total) + 3 PivotCharts

SalesData

300-row formatted Excel Table

PT_Category

Sales & Profit by Category × Region + Bar Chart

PT_YearlyComparison

2020 vs 2021 vs 2022 monthly trend + Line Chart

PT_TopProducts

Top 10 products by profit + Horizontal Bar

PT_SubCategory

All sub-categories ranked by profit (losses in red)

Slicers: Region | Category | Segment | Year | Date Timeline — all connected via Report Connections

Section 3: Power BI Dashboard — Retail Sales & Forecast BI

Data Model: Star Schema

DateTable [1] ———— [*] SalesData
Relationship: DateTable[Date] → SalesData[Order Date]
DateTable marked as official Date Table

Report Pages (4 total)

1. Overview — KPI Cards + Map + Charts
2. Category Analysis — Matrix + Scatter
3. Regional View — Filled Map + Treemap
4. Forecast — 3-month projection (95% CI)

Slicers: Date Range | Category | Region | Segment (synced across all 4 pages)

Forecasting: Analytics Pane → Forecast → 3 months ahead | Confidence: 95% | Seasonality: Auto

Export: File → Export → PDF | Save as .pbix for submission

Key DAX Measures — Power BI

Total Sales

SUM(SalesData[Sales])

$334,395

YTD Sales

TOTALYTD([Total Sales],DateTable[Date])

Cumulative

MoM Growth %

DIVIDE([Sales]-[Sales PM],[Sales PM],0)

Monthly

Profit Margin %

DIVIDE([Profit],[Sales],0)

15.4%

YoY Growth %

DIVIDE([Sales]-[Sales PY],[Sales PY],0)

+61.9%

Loss Orders

CALCULATE([Orders],Profit<0)

21 orders

Key Findings & Business Insights

Technology #1

75% of revenue — Laptops & Phones lead

2021 Best Year

YoY Growth +61.9% — highest across 3 years

West Leads

$101,195 in sales — highest region

Loss Alert

Bookcases & Tables lose money (high discounts)

Consumer King

48% revenue share — largest segment

2022 Dip

-6.8% YoY — investigate cause for recovery plan

Conclusion

This project successfully demonstrates a complete Business Intelligence workflow — from raw retail CSV data to two fully interactive dashboards using Excel and Power BI. The Excel dashboard enables Pivot Table-based KPI monitoring with interactive Slicers, while the Power BI dashboard provides advanced DAX analytics, Star Schema modeling, and 3-month sales forecasting with 95% confidence intervals.

Excel FINAL	Word Report	Power BI Guide	Dataset
Store_Sales_Dashboard_FINAL.xlsx	BEE_Lab_Report_FINAL.docx	DAX_Measures.dax + Setup Guide	retail_sales_final.csv (300 rows)

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Thank You!